AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM256
Cyber Security

Final Examination
Fall Semester 2024/2025

Date: 8/Jan./2025

Number of Exam Pages: 5 (including this cover sheet)	**Time Allowed: 3 Hours**

Instructions:

- Total Marks: **50**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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PART 1: Multiple Choice, True or False questions [15 Marks]

Question 1: Choose the correct answer for the following questions [10 Marks]:

1. Which of the following best describes a **Demilitarized Zone (DMZ)** in network?
   a) A network segment that connects all internal devices
   b) A secured area within the LAN where all critical assets are stored
   c) A buffer zone between an internal network and external networks, often used to host public-facing services
   d) A subnet dedicated to virtual private networks

2. What is **network segmentation**, and why is it commonly used in network security?
   a) Dividing a network into isolated segments to limit access to certain resources and improve security
   b) Combining multiple networks to allow unrestricted data sharing
   c) Creating multiple firewalls within a single network to restrict traffic
   d) Using virtual machines to simulate different network environments

3. Which of the following best describes **endpoint security**?
   a) Protecting only network routers and switches from unauthorized access
   b) Securing devices that connect to a network, such as computers and mobile devices, from potential threats
   c) Monitoring data traffic only within a Virtual Private Network (VPN)
   d) Isolating network segments to limit unauthorized access

4. Which of the following describes "**host defense in depth**"?
   a) Using a single security measure on each host device
   b) Isolating each host device on its own private network
   c) Installing only antivirus software on every endpoint
   d) Implementing multiple layers of security controls on a host to protect against various types of attacks

5. What is the primary goal of **application security**?
   a) To monitor network traffic and filter unauthorized access
   b) To protect applications from threats by identifying and addressing security vulnerabilities
   c) To encrypt all data in transit
   d) To install antivirus software on all network devices

6. What is the primary focus of **cloud application security (SaaS)**?
   a) Ensuring network routers are updated regularly
   b) Protecting data and applications hosted on cloud platforms
   c) Monitoring hardware resources for any unauthorized access
   d) Securing only on-premises servers and applications

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7. How does **virtualization** enhance application security?
   a) By isolating applications within virtual environments, reducing the impact of a breach on other applications
   b) By allowing unlimited data sharing between applications
   c) By encrypting all network traffic by default
   d) By automatically applying security updates

8. Which of the following describes the purpose of **DevSecOps**?
   a) To create isolated networks for different applications
   b) To install firewalls in development environments
   c) To automate security checks, ensuring security is a shared responsibility across development, security, and operations teams
   d) To conduct regular penetration tests only after application deployment

9. Which of the following is an example of a **common application security threat**?
   a) Physical tampering with network cables
   b) Unauthorized access to mobile hardware components
   c) Distributed Denial of Service (DDoS) targeting a Wi-Fi network
   d) SQL Injection, where attackers insert malicious code into a SQL query

10. How does **Cross-Site Scripting (XSS)** threaten web applications?
    a) By injecting scripts into webpages viewed by other users, potentially stealing data
    b) By making network routers less responsive
    c) By shutting down the entire web application
    d) By creating isolated environments for applications

Question 2: answer True or False to the following statements [5 Marks]:

1. Denial of Service (DoS) attacks are designed to make network services unavailable by overwhelming the target with traffic or malicious packets.
2. Reusing passwords over several systems increases the likelihood of unauthorized access to multiple accounts if one set of credentials is compromised.
3. A stream cipher encrypts one block at a time, making it faster than a block cipher.
4. Risk assessment involves understanding business operations, detailing responses, considering continuity preparations, and providing an exercise framework.
5. Discretionary Access Control (DAC) is well-suited for systems with a large number of users.

PART 2: Short Questions [20 Marks]

Question 3: Passwords are a basic element of authentication approaches. It is common to use straightforward passwords that can be guessed or hacked. Organizations often adopt a password policy to ensure that passwords provide a certain level of security. **Specify the key elements of a password policy. [4 Marks]**

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Question 4: What is a **monolithic operating system**, and what are some advantages and security concerns associated with it? **[4 Marks]**

Question 5: In recent years, as wireless networking and mobile technology have become ubiquitous in the workplace, more employees are using their own devices. Although Bring Your Own Device (BYOD) can help minimizing costs for organizations, they also introduce additional risks. **List these risks that are associated with BYOD scheme. [4 Marks]**

Question 6: What role do signatures play in antivirus systems, and what are some common methods for creating and using these signatures to identify and combat malware threats? **[5 Marks]**

Question 7: What is the purpose of using a Web Application Firewall (WAF)? **[3 Marks]**

PART 3: Problems questions [15 Marks]

Question 8: Network sniffing.

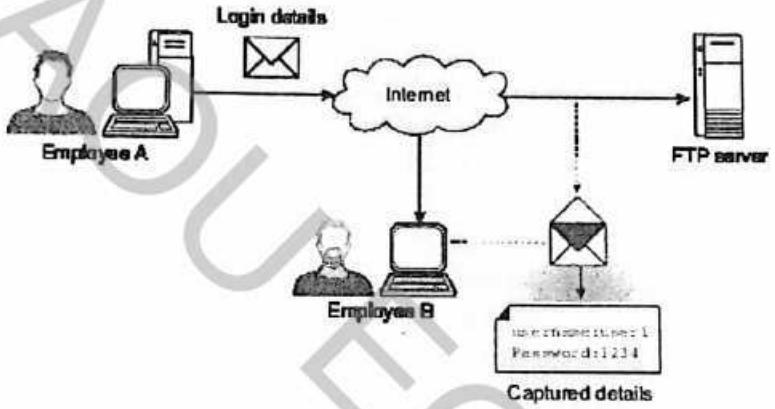


Figure 1. Network sniffing.

Figure 1 illustrates an insecure FTP server set up by Employee A, who sends login credentials over the network. Employee B, with malicious intent, uses a network sniffer to intercept the credentials sent in plaintext. Given the vulnerabilities of using insecure protocols such as FTP, answer the following questions: **[6 Marks]**

1. Explain why the FTP protocol used by Employee A is insecure in this scenario. Include in your response why protocols like FTP and Telnet are inherently insecure, especially in modern networks.
2. Suggest some security measures that the organization should implement to prevent unauthorized access and protect sensitive data.
3. Employee A needs to continue sharing files within the organization securely. Suggest an alternative protocol or method that Employee A could use instead of FTP. Explain why this alternative is more secure and how it addresses the vulnerabilities identified.

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Question 9: Exploiting a supply chain.

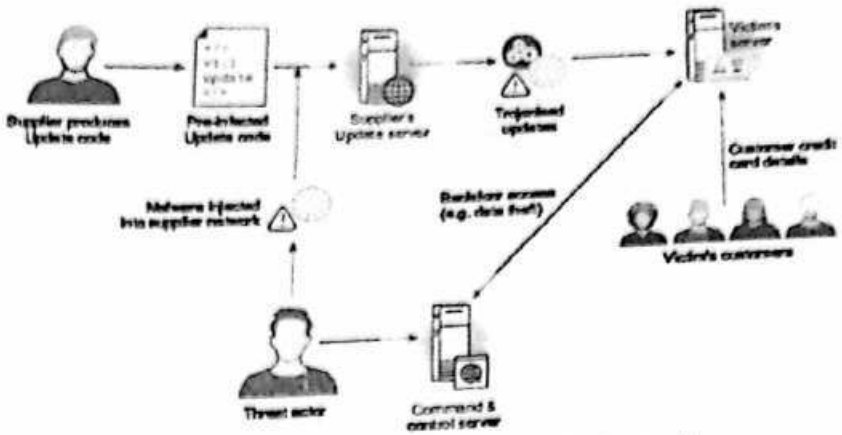


Figure 2. Software provider supply chain attack.

Figure 2 illustrates a software supply chain attack. In this scenario, the threat actor has managed to inject malware into the supplier's network, resulting in a pre-infected update code. When this code is deployed to the supplier's update server, it creates Trojanized updates that are distributed to the victim's server. These infected updates provide backdoor access to the threat actor, allowing data theft, such as customer credit card details from the victim's customers. Using Figure 2 as a reference, answer the following questions: **[5 Marks]**

1. Identify key security weaknesses in this supply chain model that the threat actor exploited. For each weakness, explain its impact on the supply chain's security.
2. Suggest mitigation strategies that could prevent such attack from occurring.

Question 10: An organization wants to build a strong security culture but finds that employees don't automatically follow secure practices. Using the "Awareness Maturity Curve", **suggest a simple step-by-step plan** to help employees move from not understanding security risks to handling them confidently and automatically. **[4 Marks]**

End of questions

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